

COIN TOSS PARADOX

LET'S CONSIDER FLIPPING HH VS FLIPPING HT.

AVERAGE NUMBER of FLIPS HH = 6

PROBABILITY HH = $1/4$

AVERAGE NUMBER of FLIPS HT = 4

PROBABILITY HT = $1/4$

A DIFFERENT PROOF THAN THE ONE ON THE SLIDES - I.E. NOT USING MARKOV CHAINS AS IN THE SLIDES

1. $X = \# \text{ FLIPS to HH}$

$$X = \frac{1}{4} \cdot 2 + \frac{1}{2} (1+X) + \frac{1}{4} (2+X)$$

$$\Rightarrow X = 6 \checkmark$$

$\frac{1}{4}$ TIMES WE GET
HH WITH THE FIRST
TWO FLIPS

$\frac{1}{4}$ TIMES WE GET HT

AND WE HAVE TO RESTART
 $\Rightarrow 2+X$ (2 FLIPS TO
HT + THE ONES WE
NEED FOR HH)

$\frac{1}{2}$ TIMES WE GET
TAILS, AND WE HAVE
TO START OVER.
 $\Rightarrow 1+X$ (ONE FLIP
WE JUST DID
+ THE ONES
WE NEED)

2. $Y = \# \text{ FLIPS to T}$

$X = \# \text{ FLIPS to HT}$

$$Y = \frac{1}{2} + \frac{1}{2} (1+Y)$$

$\frac{1}{2}$ TIMES WE GET T, $\frac{1}{2}$ TIMES WE GET H
AND WE NEED TO RESTART

$$X = \frac{1}{2} (1+X) + \frac{1}{2} (1+Y)$$

TO GET HT WE NEED TWO PIECES (H-T)
 $\frac{1}{2}$ TIMES WE GET T AND WE RESTART
 $\frac{1}{2}$ TIMES WE GET H AND WE NEED TO
GET T RIGHT AFTER (WITH
Y FLIPS)

RECURRENCE

WHAT IS THE PROBABILITY OF SEEING A SEQUENCE?
AND OF SEEING IT AGAIN?

- HTHT $\Rightarrow \frac{1}{16}$ BECAUSE WE HAVE $16 = 2^4$ POSSIBLE COMBINATIONS AND ONE INTERESTING

HOW LONG BEFORE IT APPEARS AGAIN?

16 FLIPS = , DISTANCE FROM THE START OR FROM ONE OCCURRENCE TO THE NEXT

$\Rightarrow$ THE RECURRENCE TIME FOR HTHT IS 16
(THE SAME IS FOR EVERY OTHER SEQUENCE OF LENGTH 4)

REMARK: RECURRENCE TIME $\neq$ AVERAGE TIME

EXAMPLE: HT RECURRENCE = 4
AVERAGE = 4

HH RECURRENCE = 6
AVERAGE = 6



IF WE FLIP H IN THE SECOND STATE WE START FROM THE SAME PLACE



IF WE FLIP T IN THE SECOND STATE OF THE NEXT STEP WE HAVE TO START OVER

$\Rightarrow$ RECURRENCE TIME TAKES ADVANTAGE OF THE FACT THAT WE DO NOT NECESSARILY NEED TO START FROM SCRATCH

WE CAN USE THE RECURRENCE TIME TO COMPUTE THE AVERAGE TIME

- HTHTH

$$IP(HTHTH) = \frac{1}{32}$$

RECURRENCE TIME = 32

EXPECTED TIME = 42!

[NOT ONLY THE ANSWER TO THE ULTIMATE QUESTION OF LIFE, THE UNIVERSE AND EVERYTHING]

$$S \xrightarrow{2} H \xrightarrow{8} HTH \xrightarrow{32} HTHTH = 42$$

IF WE START FROM SCRATCH WE HAVE TO "WAIT" FOR H TO OCCUR AGAIN

$$IE(HTHTH) = 32 + IE(HTH) = 32 + 8 + IE(H) = 32 + 8 + 4 = 42$$

L EXPECTED NUMBER OF FLIPS

• HTHT

$$IE[HTHT] = 16 + IE[HT] = 16 + 4 = 20$$

$$S \rightarrow \underline{HT} \rightarrow HT\underline{HT}$$

• HHTHH

$$IE[HHTHH] = 32 + IE[HH] = 32 + 6 = 38$$

(2^5)

$$S \rightarrow \underline{HH} \rightarrow HHT\underline{HH}$$

• HTTTH

$$IE[HTTTH] = 32 + IE[H] = 34$$

$$S \rightarrow \underline{H} \rightarrow H\underline{T}TTH$$

• HTTHTH

$$S \rightarrow \underline{H} \rightarrow HTTHT\underline{H}$$

↓
I cannot use this H
because if I restart from
here I get HTTTH again,
which is not what I
want